HDOC Day-45 Activity

Menu-Driven C Program - Date Format Conversion and Substrings

Name	Gurveer Singh Cheema
SAF ID	590032925
Batch	15

Aim

To write a menu-driven C program using separate functions to convert a date from dd/mm/yyyy format to dd-mmm-yy format and to print all possible contiguous substrings of a string.

Operations

1. Convert date format from **dd/mm/yyyy** to **dd-mmm-yy**.
Example: 12/10/2026 -> 12-Oct-26.
2. Print all contiguous substrings of a string.
Example: abc -> a, ab, abc, b, bc, c.
3. Exit.

Algorithm

Date conversion:
1. Read day, month and year in dd/mm/yyyy format.
2. Validate the month and basic day range.
3. Store month abbreviations in a string array.
4. Convert the four-digit year to two digits using year % 100.
5. Display dd-Mmm-yy.

All substrings:
1. Read a string.
2. Select every possible starting position.
3. For each starting position, extend to every possible ending position.
4. Print characters between the selected start and end positions.
5. Continue until all contiguous substrings are displayed.

C Program

```
#include <stdio.h>
#include <string.h>

void changeDateFormat()
{
    int day, month, year;
    const char *months[] = {
        "Jan", "Feb", "Mar", "Apr", "May", "Jun",
        "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"
    };

    printf("Enter date (dd/mm/yyyy): ");

    if (scanf("%d/%d/%d", &day, &month, &year) != 3)
    {
        printf("Invalid date format!\n");
        return;
```

```
    }

    if (month < 1 || month > 12 || day < 1 || day > 31)
    {
        printf("Invalid date!\n");
        return;
    }

    printf("Converted date: %02d-%s-%02d\n",
           day, months[month - 1], year % 100);
}

void printAllSubstrings(char str[])
{
    int i, j, k;
    int n = strlen(str);

    printf("All substrings:\n");

    for (i = 0; i < n; i++)
    {
        for (j = i; j < n; j++)
        {
            for (k = i; k <= j; k++)
                printf("%c", str[k]);

            printf("\n");
        }
    }
}

int main()
{
    int choice;
    char str[100];

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Change date format\n");
        printf("2. Print all substrings of a string\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                changeDateFormat();
                getchar();
                break;

            case 2:
                printf("Enter a string: ");
                fgets(str, sizeof(str), stdin);
                str[strcspn(str, "\n")] = '\0';

                if (strlen(str) == 0)
                    printf("String is empty!\n");
                else
                    printAllSubstrings(str);
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}
```

Sample Output

```
----- MENU -----
1. Change date format
2. Print all substrings of a string
3. Exit
Enter your choice: 1
Enter date (dd/mm/yyyy): 12/10/2026
Converted date: 12-Oct-26

Enter your choice: 2
Enter a string: abc
All substrings:
a
ab
abc
b
bc
c

Enter your choice: 3
Exiting program...
```

Function Explanation

changeDateFormat(): Reads the date as three integers separated by '/', checks the basic date range, selects the month abbreviation from the months array, and prints the last two digits of the year using **year % 100**.

printAllSubstrings(): Uses three loops. The first loop selects the starting index, the second selects the ending index, and the third prints all characters from start to end.

Easy Memory Tricks

Date: Day stays same + month number becomes name + year % 100.
12/10/2026 -> 12 + Oct + 26 -> **12-Oct-26**.

Substring pattern: Fix START -> increase END -> print START to END.
For abc: a, ab, abc | b, bc | c.

Important Note

A substring contains **continuous (contiguous)** characters from the original string. For a string of length n, the number of non-empty contiguous substrings is **n(n+1)/2**. Thus, abc has 3 x 4 / 2 = 6 substrings.

Result

Thus, the menu-driven C program was successfully implemented using separate functions to convert the date format and print all contiguous substrings of a string.

Quick Viva Questions

1. What does year % 100 give? The last two digits of the year.
2. Why is month - 1 used? C array indexing starts from 0, while month numbers start from 1.
3. What is a substring? A continuous sequence of characters taken from a string.
4. How many non-empty substrings does a string of length n have? n(n+1)/2.
5. How many substrings does abc have? 6.
6. What are the substrings of abc? a, ab, abc, b, bc, c.